Problem 1: You are given the value of n, you need to find the result of $\sum_{i=1}^{n} i*(n-i+1)$. Solution should be O(1) and can contain if-else but no iteration.

(Mahbubul, Programming Contest: Data Structure and Algorithm, 2016. Page 28)

Solution:

$$\sum_{i=1}^{n} i(n-i+1) = \sum_{i=1}^{n} (ni - i^2 + i)$$

$$= \sum_{i=1}^{n} (ni) - \sum_{i=1}^{n} (i^2) + \sum_{i=1}^{n} (i)$$

$$= n\sum_{i=1}^{n} (i) + \sum_{i=1}^{n} (i) - \sum_{i=1}^{n} (i^2)$$

$$= (n+1)\sum_{i=1}^{n} (i) - \sum_{i=1}^{n} (i^2)$$

$$= (n+1)\left(\frac{n(n+1)}{2}\right) - \left(\frac{n(n+1)(2n+1)}{6}\right)$$

$$= n(n+1)\left(\frac{n+1}{2} - \frac{2n+1}{6}\right)$$

$$= n(n+1)\left(\frac{n+2}{6}\right)$$

Pre-assumptions:

1. Summation rules for closed-form expressions are known. Which are:

(a) $\sum_{k=1}^{n} (c) = nc$

(b) $\sum_{i=1}^{n} (c \cdot i) = c \sum_{i=1}^{n} (i)$

(c) $\sum_{i=1}^{n} (x_i \pm y_i) = \sum_{i=1}^{n} (x_i) + \sum_{i=1}^{n} (y_i)$

2. Summation formulas for close-form expressions are known. Which are:

(a) $\sum_{i=1}^{n} (i) = \frac{n(n+1)}{2}$

(b) $\sum_{i=1}^{n} (i^2) = \frac{n(n+1)(2n+1)}{6}$

3. There is no familiarity with pascal's triangle! We don't about it! So no shortcuts from there!